

Utsav Acharya

✉ clerisy47@gmail.com ☎ +9779860307243 📄 Portfolio 🌐 LinkedIn 🏠 Github

An ever learning **Computer Engineering** student with a strong foundation in **Mathematics**, **Data Science**, **Machine Learning**, and **Backend Development** fundamentals and person with an urge for continual learning, problem solving and innovation.

WORK EXPERIENCE

Student Trainee 04/2025 – 11/2025
NAAMI

Conducting computer vision research in the physics lab focused on:

- **Disease detection** in rice crops
- **Behavioral monitoring** of poultry

Backend Engineer and Data Scientist 11/2023 – 03/2025
Omdena 📄

Participated in the following real-world deployed projects:

- **Combatting illegal deforestation in Romania** 📄
- **Mapping Seagrass Meadows** 📄
- **Agriculture Analysis on Kenya** 📄

Data Science and Business Analytics Intern 09/2023 – 10/2023
The Sparks Foundation

- Created raspberry pi app for **Tomato Disease Classification** 📄
- Did **Exploratory Data Analysis on IPL 2023** 📄

ACADEMIC PROJECTS

MEDIWO — LLM-Powered Medical Workflow Optimization 📄

An AI-assisted platform that optimizes OPD workflows by automating patient data collection during waiting times and delivering structured clinical summaries to doctors before consultation.

- Used React, TypeScript and Tailwind CSS for frontend with separate patient and doctor dashboards
- Used **FastAPI** and **Python** for backend with MongoDB for patient records and Chroma as vector database
- Implemented **RAG pipeline** using Gemini 2.5 Pro and **LangChain** for clinical summarization
- Built adaptive **AI intake chatbot** for pre-consultation symptom collection
- Integrated **OCR module** for digitizing and extracting data from uploaded medical reports
- Implemented **Human In the Loop** feedback loop logging doctor edits as reward signals for continuous model improvement

Compliance Guide 📄

The Compliance Guide saves you countless hours by simplifying complex policy documents and compliance frameworks, helping you ensure seamless security and a gap-free product.

- Used **Llama** for summariser
- Increased accuracy to upto 95% using **RAG**
- Used locally hosted **Llama** as comparator
- Used **Streamlit** for creating dashboards

Volt Vision 📄

Energy demand forecasting for Nepal households.

- Used pandas for data analysis and **Xgboost** for demand forecasting.
- Used **Sarimax** and **Arimax** model for time series analysis and price forecasting.
- Deployment Pipelines for Energy Demand Forecasting System using ZenML/MLflow

Guardian Shield 📄

Guardian-Shield is a browser extension designed to enhance your online security by detecting and preventing common fraud and phishing attempts in your emails.

- Used **Chrome apis** for extension
- Used **Bert Transformer** and **Pytorch** for spam detection
- Use **Peagsus Transformer** and pytorch for terms and conditions summarizing.

EDUCATION

Bachelor's in Computer Engineering

Pulchowk Engineering Campus

Average Percentage: 82.4

2023 – present
Kathmandu, Nepal

High School

Capital College and Research Centre

GPA: 3.91 + 3.73

2021 – 2023
Kathmandu, Nepal

SKILLS

Languages — Python, C, C++, JavaScript

Frameworks — Numpy, Pandas, Flask, Fastapi, Django, Matplotlib, Seaborn, Scikit Learn, TensorFlow, OpenCV, spaCy, React

Developer Tools — Git, Docker, VS Code, Tableau

AWARDS

National IT Olympiad Silver Medalist

STEM, Nepal

2022

National Hackathon Winner

LOCUS

2024

Runner UP:- Nepal Hacks

Security Pal, NAAMI

2024

Specialist, National top 10

Codeforces

2024

Tribhuvan University Merit-Based Scholarship

Tribhuvan University

2023

National Physics-Mathematics Olympiad Bronze Medalist

SOMES Nepal

2021

CERTIFICATES

- | | | |
|---|---|--|
| • Google Advanced Data Analytics 🔗 | • IBM Data Science Professional Certificate 🔗 | • Stanford Machine Learning Specialization 🔗 |
| • Tensorflow Developer Professional Certificate 🔗 | • Meta Backend Professional Certificate 🔗 | |

LEADERSHIP & EXTRACURRICULARS

Data Research Council For Students

Core Member

11/2022 – 12/2023

- Mentored **200+** students in **intermediate Python**, providing guidance and support to help them improve their skills.
- Managed **GitHub** repositories, **social media** accounts, and **Discord** servers, ensuring that these platforms were up-to-date and effectively engaged with our community.

Together We Learn

Co founder

03/2022 – 08/2023

- Co-founded and led an organization of **500+** members
- Conducted **5+** **webinars** on programming-related topics, sharing knowledge and insights with a broad audience.
- Wrote **10+** **newsletters** focused on **data science** and **programming** in general, providing valuable information and updates to **200+** **subscribers**.

Kathmandu Valley Leo Club

Volunteer

01/2023 – 07/2025

- Participated in different volunteering activities from KVLC Pulchowk.